

Assignment 7: Microprogrammed Control Unit Design

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Deadline: Friday, 21 August 2026, 11:59 PM — exactly *two weeks* from issue (07 August 2026). There is no automatic late penalty — if you need more time, ask before the deadline and an extension will be granted (see the course policy in the syllabus).

Answer every question. Show all working for Numerical and Design questions.

7.1 Conceptual

Q1. Explain the role of the mapping unit in a microprogrammed control unit. What would happen if the mapping unit were removed and MPC simply reset to address 0 after every END?

Q2. Compare horizontal and vertical microinstruction encoding in terms of (a) control-word width and (b) the delay before control signals become valid after CM is read.

7.2 Numerical

Q3. A machine has 20 mutually-exclusive bus-source signals (exactly one is active at a time). How many bits are needed to encode this as a single vertical field? How many bits would the same 20 signals need if encoded horizontally (one bit per signal)?

Q4. ADD requires 3 control-memory rows (as in the lecture’s worked example) and SUB requires a similarly-shaped 3-row microprogram. A third instruction, a register-only NOT R1 (one-operand, no memory or bus-buffering needed beyond reading and writing R1), requires 2 rows. Assuming no rows are shared between instructions, how many total control-memory rows are needed to hold all three microprograms?

7.3 Design

Q5. Write the microprogram (microaddress, RTL, control word, next-address field) for $R4 \leftarrow R5 - R6$ on the single-bus datapath, in the same table format as the lecture’s worked example for $R1 \leftarrow R2 + R3$. Your microprogram must have 3 rows, correct next-address fields, and an END marker on the last row.

Q6. The worked ADD microprogram in the lecture uses 8 distinct control signals: $R2_{out}$, $R3_{out}$, Y_{in} , $ALU=add$, Z_{in} , Z_{out} , $R1_{in}$, END. Design a vertical encoding scheme for these 8 signals by grouping mutually-exclusive signals into encoded fields (following the lecture’s worked example, which grouped bus-source, destination, and ALU-operation into three fields). State your field groupings and the resulting total control-word width in bits, compared to the 8 bits horizontal encoding would require.